

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$x + y = 350$

The given equation relates the total number of maple trees, x , and the total number of birch trees, y , planted in a **14**-acre forest preserve. If **245** maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

Answer

- A. **14**
- B. **25**
- C. **105**
- D. **245**

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equation $x + y = 350$ relates the total number of maple trees, x , and the total number of birch trees, y , planted in a **14**-acre forest preserve. It’s also given that **245** maple trees were planted in the forest preserve. Substituting **245** for x in the given equation yields $245 + y = 350$. Subtracting **245** from both sides of this equation yields $y = 105$. Therefore, **105** birch trees were planted in the forest preserve.

Choice A is incorrect. This is the number of acres in the forest preserve, not the number of birch trees planted in the forest preserve.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the number of maple trees planted in the forest preserve, not the number of birch trees planted in the forest preserve.